Elaboration of the Scale Selection Pipeline for Autoregressive Processes

Prepared notes for the thesis chapter

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Abstract

These notes expand the proposed application of the spectral scale-selection pipeline to autoregressive processes. The main idea is to replace the usual filtration parameter ε by the embedding dimension k, or to study the two parameters (k, ε) jointly. For an AR(p) process, delay vectors of length k have a distinctive geometric structure: for $k = p + 1$ they concentrate near one autoregressive hyperplane; for $k > p + 1$ they satisfy several shifted autoregressive slab constraints; and the effective dimension stops increasing after the relevant autoregressive memory has been included. This suggests several ways of selecting p spectrally: through intrinsic-dimension saturation, prediction-residual functionals, stability of Vietoris–Rips filtrations, persistence-diagram similarities, zigzag persistence, and time-preserving temporal complexes. The most promising thesis-level route is a two-stage procedure: first use ε-filtrations at each k to extract a robust critical scale or geometry score, then apply the existing Perron–Frobenius scale-selection principle to the resulting sequence over k.

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1 Starting point and conceptual goal

Your current scale-selection framework starts from a finite ordered family of complexes, usually a filtration

$$K_{\varepsilon_1}(X) \subseteq K_{\varepsilon_2}(X) \subseteq \cdots \subseteq K_{\varepsilon_N}(X),$$

evaluates a nonnegative functional f along this family, and forms a scale-similarity matrix whose entries encode the interaction of two scales. In the simplest filtration case one obtains a min/meet-type matrix

$$\Sigma_{ij} = f(K_{\varepsilon_i \wedge \varepsilon_j}), \qquad i \wedge j = \min\{i, j\}.$$

The Perron–Frobenius eigenvector v^* then ranks the scales globally rather than only through pointwise local extrema. In the autoregressive application, the conceptual change is:

$$\text{filtration scale } \varepsilon \quad \rightsquigarrow \quad \text{embedding dimension } k.$$

Thus the state space of the spectral selection problem becomes

$$\{1, 2, \ldots, k_{\max}\}$$

instead of a grid of radii $\{\varepsilon_1, \ldots, \varepsilon_N\}$.

The mathematical target is not merely to build a topological summary of a time series. The target is sharper: construct a triple

$$(f, \{K_k\}_{k=1}^{k_{\max}}, \oplus)$$

which is spectrally optimizable and whose selected index

$$k_{\text{crit}} \in \operatorname*{arg\,max}_{1 \le k \le k_{\max}} v_k^*$$

contains information about the autoregressive order p. Since an AR(p) process has memory p, the most natural expectations are either

$$k_{\text{crit}} \approx p, \qquad \text{or} \qquad k_{\text{crit}} \approx p + 1.$$

Which of these two is correct depends on the convention. If $Y_t^{(k)} = (X_t, \ldots, X_{t-k+1})$, then the first embedding dimension in which the autoregressive equation becomes a codimension-one relation among coordinates is $k = p + 1$. Therefore a geometrically motivated procedure will often detect $p + 1$, from which the order is estimated by $\hat{p} = k_{\text{crit}} - 1$. A prediction-based procedure may instead detect p directly, because p past coordinates are enough to predict the next value.

2 Autoregressive geometry of sliding-window point clouds

Let (X_t) be a stationary AR(p) process

$$X_t = \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + Z_t, \qquad \phi_p \neq 0, \tag{1}$$

where (Z_t) is an i.i.d. innovation sequence. For fixed k, define the delay vector

$$Y_t^{(k)} = (X_t, X_{t-1}, \ldots, X_{t-k+1}) \in \mathbb{R}^k, \qquad \mathcal{P}_k = \{Y_t^{(k)} : t = k, \ldots, T\}.$$

This is the standard delay-coordinate construction used in dynamical systems and time-series TDA. The classical deterministic reference point is Takens' theorem [1], together with its extensions such as Sauer–Yorke–Casdagli's embedology framework [2]. In applied nonlinear time-series analysis, the delay and embedding dimension are often chosen by mutual information [3] and false nearest neighbors [4]. Your construction differs from these classical approaches because the candidate dimensions are not only assessed geometrically, but are passed through a spectral scale-selection principle.

2.1 Why forward and backward embeddings are essentially equivalent

One may use

$$Y_t^{(k),-} = (X_t, X_{t-1}, \ldots, X_{t-k+1})$$

or

$$Y_t^{(k),+} = (X_t, X_{t+1}, \ldots, X_{t+k-1}).$$

For a strictly stationary process, both have the same distribution up to coordinate reversal and a shift of the time index, provided the finite-dimensional distributions are time reversible or the chosen statistic is invariant under the induced coordinate permutation. Even without full time reversibility, the Vietoris–Rips complex built from Euclidean distances is invariant under an orthogonal coordinate reversal. Thus, for stationary AR processes, the geometry of the cloud is not fundamentally changed by using past or future windows. What may change is the interpretation: the backward embedding is naturally prediction-oriented, whereas the forward embedding is naturally constraint-oriented.

2.2 The case $k = p + 1$: one autoregressive slab

For $k = p + 1$, the coordinates of

$$Y_t^{(p+1)} = (X_t, X_{t-1}, \ldots, X_{t-p})$$

satisfy

$$X_t - \phi_1 X_{t-1} - \cdots - \phi_p X_{t-p} = Z_t. \tag{2}$$

If the innovations are centered, the cloud is centered around the hyperplane

$$H_0 = \{x \in \mathbb{R}^{p+1} : x_0 - \phi_1 x_1 - \cdots - \phi_p x_p = 0\}.$$

If $Z_t \sim U[-a, a]$, then the support is contained in the slab

$$\{x \in \mathbb{R}^{p+1} : |x_0 - \phi_1 x_1 - \cdots - \phi_p x_p| \le a\}.$$

This is the first major geometric signature of the AR order: at $k = p + 1$ the delay cloud becomes organized around a codimension-one autoregressive relation.

2.3 The case $k > p + 1$: multiple slabs and dimension saturation

For $k > p + 1$, each delay vector satisfies several shifted versions of the AR equation:

$$x_j - \phi_1 x_{j+1} - \cdots - \phi_p x_{j+p} = Z_{t-j}, \qquad j = 0, \ldots, k - p - 1, \tag{3}$$

where x_j denotes the j-th coordinate of $Y_t^{(k)}$. For bounded innovations $Z_t \in [-a, a]$, the cloud lies in the intersection of $k - p$ slabs:

$$\bigcap_{j=0}^{k-p-1} \{x \in \mathbb{R}^k : -a \le x_j - \phi_1 x_{j+1} - \cdots - \phi_p x_{j+p} \le a\}.$$

The central intersection is

$$L_k = \bigcap_{j=0}^{k-p-1} \{x \in \mathbb{R}^k : x_j - \phi_1 x_{j+1} - \cdots - \phi_p x_{j+p} = 0\}.$$

For generic stable AR coefficients, this central object has dimension p: once $x_{k-p}, \ldots, x_{k-1}$ are known, the remaining coordinates are recursively determined. Hence the noise-free deterministic recurrence lives on a p-dimensional linear subspace in every ambient dimension $k \ge p + 1$.

This gives a strong thesis idea:

Principle 1 (Dimension saturation principle)**.** *For an AR(p) process, the effective dimension of the delay cloud should increase with k until the memory length has been reached and should then stabilize. A spectral procedure over k should therefore be designed to detect the transition between increasing effective dimension and saturation.*

This principle is close in spirit to false nearest neighbors: when k is too small, points that are close in $\mathbb{R}^k$ may separate after adding an additional lag; once k is large enough, neighborhoods stabilize [4]. Your contribution can be framed as a topological/spectral analogue of this idea.

2.4 Support geometry through the MA(∞) representation

If the AR process is causal and stationary, it has an MA(∞) representation

$$X_t = \sum_{\ell=0}^{\infty} \psi_\ell Z_{t-\ell},$$

where $\psi_0 = 1$ and

$$\psi_\ell = \phi_1\psi_{\ell-1} + \cdots + \phi_p\psi_{\ell-p}, \qquad \psi_{-1} = \cdots = \psi_{-p} = 0.$$

Causality implies exponential decay of $|\psi_\ell|$ when the roots of the AR polynomial lie outside the unit disk; see standard ARMA theory, e.g. Brockwell–Davis [5]. For $k = p + 1$,

$$Y_t^{(p+1)} = \sum_{j=0}^{\infty} g_j Z_{t-j}, \qquad g_j = (\psi_j, \psi_{j-1}, \ldots, \psi_{j-p})^\top.$$

If $Z_t \in [-a, a]$, then

$$\operatorname{supp}(Y_t^{(p+1)}) \subseteq \sum_{j=0}^{\infty} [-a, a] g_j,$$

a centrally symmetric zonoid. Its finite truncations

$$Z_n = \sum_{j=0}^{n} [-a, a] g_j$$

are zonotopes. Since $\sum_j \|g_j\| < \infty$, the Hausdorff distance between the full zonoid and the truncation is bounded by

$$d_H\left(\sum_{j=0}^{\infty} [-a, a] g_j, \sum_{j=0}^{n} [-a, a] g_j\right) \le a \sum_{j=n+1}^{\infty} \|g_j\|.$$

This gives a clean way to make your geometric discussion rigorous: stability of the AR process implies exponential decay of the tail, hence fast approximation by finite zonotopes.

2.5 Effect of innovation distribution

The innovation law changes the cloud geometry substantially.

- **Bounded innovations.** Uniform or bounded innovations produce slab-bounded clouds and compact zonotopal support. This is the easiest case for figures and for geometric proofs.
- **Gaussian innovations.** The delay vector is multivariate Gaussian. The cloud no longer has bounded support; instead it is concentrated near ellipsoids determined by the Toeplitz covariance matrix $\Gamma_k = (\gamma(i-j))_{i,j=0}^{k-1}$. In this case one should replace hard slabs by high-probability slabs, for example $|x_0 - \sum_i \phi_i x_i| \le c\sigma_Z$.
- **Heavy-tailed innovations.** Student-t innovations create outliers that can dominate Vietoris–Rips complexes. This is useful experimentally: a robust version of the pipeline should either use distance truncation, k-nearest-neighbor complexes, witness complexes, or robust functionals.

3 Three ways to apply the scale-selection pipeline

There are three conceptually different ways to apply your method.

3.1 Route A: k as a hyperparameter, ε selected inside each k

For each $k = 1, \dots, k_{\max}$, construct the usual filtration

$$\mathrm{VR}_{\varepsilon_1}(\mathcal{P}_k) \subseteq \cdots \subseteq \mathrm{VR}_{\varepsilon_N}(\mathcal{P}_k).$$

Apply your existing scale-selection pipeline to obtain

$$\varepsilon_{\mathrm{crit}}(k), \qquad v^{*,k}, \qquad \lambda^*(k),$$

and then analyze the profiles

$$k \mapsto \varepsilon_{\mathrm{crit}}(k), \qquad k \mapsto \lambda^*(k), \qquad k \mapsto \max_j v_j^{*,k}.$$

This is the most robust route because it does not force complexes in different ambient dimensions to be nested. It respects the fact that $\mathcal{P}_k \subset \mathbb{R}^k$ and $\mathcal{P}_k + 1 \subset \mathbb{R}^{k+1}$ are different point clouds, not a filtration in the ordinary sense.

A strong order-estimation rule is:

$$\hat{p} = \arg\max_k \Delta_k G(k),$$

where $G(k)$ is a geometry score extracted from the ε-filtration and $\Delta_k G(k) = G(k+1) - G(k)$. Candidate scores include:

$$G_1(k) = \varepsilon_{\mathrm{crit}}(k), \quad G_2(k) = \lambda^*(k), \quad G_3(k) = \sum_j v_j^{*,k} \varepsilon_j, \quad G_4(k) = \max_j f(\mathrm{VR}_{\varepsilon_j}(\mathcal{P}_k)).$$

One expects a visible transition around p or $p+1$.

3.2 Route B: build a similarity matrix directly over dimensions

Instead of forcing a filtration over k, compute a signature S_k for each point cloud $\mathcal{P}_k$ and compare signatures:

$$M_{k\ell} = \kappa(S_k, S_\ell).$$

Then apply Perron–Frobenius to M. This is a direct extension of your scale-similarity philosophy from scales to embedding dimensions. Several choices are natural:

- $S_k = D_k$, the persistence diagram of $\mathcal{P}_k$, and

$$M_{k\ell} = \exp\left(-\frac{d(D_k, D_\ell)^2}{2\sigma^2}\right),$$

  where d is bottleneck or Wasserstein distance. Persistence-diagram kernels and stable vectorizations are well studied; see persistence images [12], persistence landscapes [10], and persistence scale-space kernels [11].

- S_k is a vector of scale-selected features: critical scale, Betti numbers at the critical scale, edge density, clustering coefficient, local dimension estimate, prediction residual.

- S_k is the covariance spectrum of $\mathcal{P}_k$. This gives a linear baseline comparable to PCA and classical AR order diagnostics.

This route avoids the artificial requirement $K_1 \subseteq \cdots \subseteq K_{k_{\max}}$.

3.3 Route C: create an actual filtration over k

Your notes propose constructing complexes K_k on a common vertex set

$$V = \{k_{\max}, \dots, T\}$$

by adding simplices along consecutive time indices:

$$K_k = \langle \{\{r, \dots, r+j\} : 0 \le j \le k-1,\ r = k_{\max}, \dots, T-j\} \rangle .$$

This indeed gives

$$K_1 \subseteq K_2 \subseteq \cdots \subseteq K_{k_{\max}}.$$

However, as stated, this construction depends only on time order and not on the geometry of the lifted clouds $\mathcal{Q}_k$. Therefore it cannot recover the AR order unless the functional f also uses geometry, weights, prediction errors, or distances.

The fix is to use a *weighted temporal complex*. Define a simplex $\sigma = \{r, \dots, r+j\}$ to have a weight depending on the lifted points, for example

$$w_k(\sigma) = \operatorname{diam}\{p_r^k, \dots, p_{r+j}^k\},$$

or

$$w_k(\sigma) = \sum_{s=r}^{r+j-1} \|p_{s+1}^k - p_s^k\|.$$

Then one obtains a bifiltration-like object indexed by (k, ε):

$$K_{k,\varepsilon} = \{\sigma \in K_k : w_k(\sigma) \le \varepsilon\}.$$

This is more faithful than the unweighted construction because k controls temporal order and ε controls geometry.

4 Dimension-lifting strategies

Your notes ask how to lift all $\mathcal{P}_k \subset \mathbb{R}^k$ into a common $\mathbb{R}^{k_{\max}}$. This is important if one wants to compare point clouds at different k using the same ambient metric.

4.1 Zero padding

The simplest lift is

$$\iota_k(X_t, \dots, X_{t-k+1}) = (X_t, \dots, X_{t-k+1}, 0, \dots, 0) \in \mathbb{R}^{k_{\max}}.$$

This is easy but not ideal: the missing coordinates are not truly zero, and distances between different k may be distorted by the arbitrary origin.

4.2 Mean padding

A small improvement is to pad with the empirical mean $\bar{X}$:

$$\iota_k(Y_t^{(k)}) = (X_t, \dots, X_{t-k+1}, \bar{X}, \dots, \bar{X}).$$

For centered stationary processes this coincides with zero padding, but it is more robust when the mean is not zero.

4.3 Linear prediction padding

Your proposed lift is substantially better. Write

$$q_t^k = (X_t, \dots, X_{t-k+1}), \qquad z_t^k = (X_{t-k}, \dots, X_{t-k_{\max}+1}).$$

Estimate

$$(\hat{A}_k, \hat{b}_k) \in \arg\min_{A,b} \sum_{r=k_{\max}}^{T} \|z_r^k - Aq_r^k - b\|^2$$

and set

$$p_t^k = (q_t^k, \hat{A}_k q_t^k + \hat{b}_k) \in \mathbb{R}^{k_{\max}}.$$

This lift has a clear interpretation: p_t^k is the best linear reconstruction of the full delay vector using only the first k observed coordinates. For AR processes, this is especially meaningful because linear prediction is the correct prediction class. The reconstruction error

$$R(k) = \frac{1}{T - k_{\max} + 1} \sum_{t=k_{\max}}^{T} \|z_t^k - \hat{A}_k q_t^k - \hat{b}_k\|^2$$

should decrease with k and show an elbow near p. A nonmonotone version suitable for your spectral pipeline is

$$F_{\text{pred}}(k) = \big(R(k-1) - R(k)\big)\,\big(R(k) - R(k+1)\big)_+^{-1},$$

or more stably

$$F_{\text{curv}}(k) = \left|R(k+1) - 2R(k) + R(k-1)\right|.$$

These quantities turn the classical elbow idea into a functional over k.

4.4 Conditional-expectation padding

For Gaussian AR processes, the optimal L^2 lift is the conditional expectation

$$\hat{z}_t^k = \mathbb{E}[z_t^k \mid q_t^k].$$

Because the joint distribution is Gaussian, this conditional expectation is linear and can be written using covariance blocks:

$$\hat{z}_t^k = \mu_z + \Gamma_{zq}\Gamma_{qq}^{-1}(q_t^k - \mu_q).$$

This is the population version of the regression lift. It gives a theoretical benchmark for simulations.

4.5 Projection from the full cloud

Another option is to regard the full cloud

$$\mathcal{P}_{k_{\max}} = \{Y_t^{(k_{\max})}\}$$

as the master object and define $\mathcal{P}_k$ as its projection onto the first k coordinates. Then one does not lift lower-dimensional clouds; instead one studies the loss of information under projection. This connects directly to false nearest neighbors: if projection to $\mathbb{R}^k$ creates artificial neighborhoods that disappear in $\mathbb{R}^{k+1}$, then k is too small.

5 Time-preserving complexes

A central issue in your notes is that ordinary Vietoris–Rips complexes ignore temporal order. For AR processes this is dangerous: the same geometric cloud can arise from different temporal traversals. Several constructions retain temporal information.

5.1 Temporal path complex

Your consecutive-index complex is a path-based construction. It is related in spirit to path complexes and directed/ordered simplicial structures used for networks and time-ordered data. The simplex

$$\{r, r+1, \ldots, r+j\}$$

records a contiguous time block. The maximal dimension $k-1$ says how long a block is allowed to be. This is a good way to make k itself into a filtration parameter, but it should be weighted by geometry or prediction error.

5.2 Vietoris–Rips complex with temporal edge filter

Define an edge between s and t if both

$$\|p_s^k - p_t^k\| \le \varepsilon \quad \text{and} \quad |s-t| \le L.$$

Then take the flag complex. The parameter L controls temporal locality. If $L = 1$, one recovers a path-like complex. If L is large, one approaches ordinary Vietoris–Rips. This construction is attractive because it interpolates between purely temporal and purely geometric structure.

5.3 Directed temporal complex

For a time series, the order $t \to t+1$ is meaningful. One may define directed edges

$$t \to s \quad \text{if} \quad 0 < s - t \le L \text{ and } \|p_s^k - p_t^k\| \le \varepsilon.$$

Directed flag complexes or path homology can then be used. This is more advanced and may be too much for a master's thesis, but it provides a research-level extension.

5.4 Witness complex with temporal landmarks

Choose landmarks along time, for example every m-th delay vector or points selected by max-min sampling in the lifted cloud. Build a witness complex using all delay vectors as witnesses. Temporal information can be retained by selecting landmarks in time order or by weighting witnesses according to temporal distance. Witness complexes are computationally cheaper than full Vietoris–Rips complexes and are standard in TDA.

6 Operations $\oplus$ on complexes

Your framework asks how to combine K_k and K_ℓ.

6.1 Union

The union

$$K_k \oplus K_\ell = K_k \cup K_\ell$$

is the most natural choice when the family is nested. If $K_k \subseteq K_\ell$ for $k \le \ell$, then

$$K_k \cup K_\ell = K_\ell, \qquad K_k \cap K_\ell = K_k.$$

This recovers min/meet-matrix behavior.

6.2 Intersection

The intersection emphasizes features common to two dimensions:

$$K_k \oplus K_\ell = K_k \cap K_\ell.$$

For non-nested geometric complexes, this can identify stable simplices that survive changes in embedding dimension.

6.3 Common-neighbor or pullback operation

If all complexes are built on the same vertex set V, define

$$K_k \wedge K_\ell = \{\sigma \subseteq V : \sigma \in K_k \text{ and } \sigma \in K_\ell\}.$$

This is intersection, but interpreted as a common-neighborhood operation. It asks: which temporal blocks or geometric relations are visible at both dimensions?

6.4 Zigzag operation

If there is no honest filtration, use a zigzag:

$$K_1 \to K_1 \cup K_2 \leftarrow K_2 \to K_2 \cup K_3 \leftarrow K_3 \to \cdots.$$

Zigzag persistence was introduced precisely to handle sequences of spaces that are related by maps but not nested [7]. This is a principled way to compare K_k across k without pretending that $K_k \subseteq K_{k+1}$.

7 Functional choices

The functional is the most important design choice. It should be sensitive to the AR order and should not be trivially monotone unless you use your monotone variant.

7.1 Prediction-residual functional

Define

$$R(k) = \frac{1}{n} \sum_t \|Y_t^{(k_{\max})} - p_t^k\|^2.$$

This is monotone decreasing in k for nested linear models. To make it suitable for spectral selection, use either improvement

$$I(k) = R(k-1) - R(k),$$

or curvature

$$C(k) = R(k+1) - 2R(k) + R(k-1).$$

For an AR(p) process, the marginal prediction improvement should drop after p relevant lags. Therefore $I(k)$ or $|C(k)|$ should peak near p.

7.2 False-nearest-neighbor functional

Let $NN_k(t)$ be the nearest neighbor of $Y_t^{(k)}$ in $\mathbb{R}^k$. Define

$$F_{\text{FNN}}(k) = \frac{1}{n} \sum_t \mathbf{1}\left\{ \frac{|X_{t-k} - X_{NN_k(t)-k}|}{\|Y_t^{(k)} - Y_{NN_k(t)}^{(k)}\|} > \tau \right\}.$$

This is the classical false-nearest-neighbor idea [4]. It is usually decreasing. A spectral version can select the point where the decrease stabilizes, for example by applying the monotone scale-selection variant to $1 - F_{\text{FNN}}(k)$ or by using the curvature of F_{FNN}.

7.3 Intrinsic-dimension functional

Let $d_{\mathrm{loc}}(k)$ be an estimated intrinsic dimension of $\mathcal{P}_k$, for example from local PCA, correlation dimension, or nearest-neighbor scaling. Then use

$$F_{\dim}(k) = d_{\mathrm{loc}}(k+1) - d_{\mathrm{loc}}(k)$$

or

$$F_{\mathrm{sat}}(k) = \left| d_{\mathrm{loc}}(k+1) - 2d_{\mathrm{loc}}(k) + d_{\mathrm{loc}}(k-1) \right|.$$

For AR(p), the dimension should saturate around p or $p+1$ depending on noise and convention.

7.4 Topological-combinatorial functional

For each k and ε, compute

$$f_{\alpha,\gamma}(K_{k,\varepsilon}) = \beta_0(K_{k,\varepsilon})^{\alpha} f_1(K_{k,\varepsilon})^{\gamma}.$$

This is the same type of nonmonotone product used in your general pipeline: β_0 decreases as edges connect components, while f_1 increases as edges are added. For each k, select $\varepsilon_{\mathrm{crit}}(k)$ and define

$$G(k) = f_{\alpha,\gamma}(K_{k,\varepsilon_{\mathrm{crit}}(k)}).$$

Then apply a second spectral selection over k. This two-level pipeline is clean and thesis-friendly.

7.5 Autoregressive slab functional

If AR coefficients are estimated for each candidate order k, define residuals

$$\hat{Z}_t^{(k)} = X_t - \hat{\phi}_1^{(k)} X_{t-1} - \cdots - \hat{\phi}_k^{(k)} X_{t-k}.$$

A geometric functional can measure slab thickness:

$$S(k) = \mathrm{MAD}\{\hat{Z}_t^{(k)}\}$$

or the entropy of residuals. Since overfitting can always reduce residual variance, one should penalize dimension:

$$F_{\mathrm{slab}}(k) = \log S(k) + \eta k / T.$$

This connects your method to classical information criteria, but the selected k is obtained by the spectral principle rather than by AIC/BIC alone.

7.6 Persistence-diagram stability functional

Compute persistence diagrams D_k of $\mathcal{P}_k$. Define a local change score

$$\Delta_D(k) = d(D_k, D_{k+1}).$$

One expects large changes before the correct embedding dimension and smaller changes afterward. Therefore use

$$F_D(k) = \Delta_D(k-1) - \Delta_D(k)$$

or construct a full kernel matrix

$$M_{k\ell} = \exp\left(-d(D_k, D_\ell)^2 / (2\sigma^2) \right).$$

This is strongly supported by the literature on persistence-diagram distances and kernels [9, 11, 12].

8 A recommended thesis-level pipeline

The most defensible pipeline is the following.

Step 1: simulate or observe a time series

Generate stationary AR(p) processes for known p and multiple innovation laws: Gaussian, uniform, and Student-t. Standardize the series.

Step 2: build delay clouds

For $k = 1, \dots, k_{\max}$, construct

$$\mathcal{P}_k = \{(X_t, \dots, X_{t-k+1}) : t = k, \dots, T\}.$$

Optionally subsample to reduce serial dependence.

Step 3: compute a filtration for each k

Build $\mathrm{VR}_\varepsilon(\mathcal{P}_k)$ over a grid of ε values. To control high-dimensional distance concentration, use standardized coordinates or a correlation distance variant.

Step 4: select an internal scale

Use your current Perron–Frobenius method to select $\varepsilon_{\mathrm{crit}}(k)$ from a nonmonotone functional such as $\beta_0 f_1$.

Step 5: create a sequence over k

Define a dimension score

$$G(k) = f(\mathrm{VR}_{\varepsilon_{\mathrm{crit}}(k)}(\mathcal{P}_k)),$$

or combine several scores:

$$G(k) = w_1 G_{\mathrm{top}}(k) + w_2 G_{\mathrm{pred}}(k) + w_3 G_{\mathrm{dim}}(k).$$

Normalize each component before combining.

Step 6: apply spectral selection over k

Construct

$$\Sigma_{k\ell} = G(k \wedge \ell) + \delta,$$

where $\delta > 0$ ensures positivity, or use a kernel matrix $M_{k\ell} = \exp(-\|S_k - S_\ell\|^2/(2\sigma^2))$. Let v^* be the Perron–Frobenius eigenvector and set

$$k_{\mathrm{crit}} = \arg\max_k v_k^*.$$

Estimate either $\hat{p} = k_{\mathrm{crit}}$ or $\hat{p} = k_{\mathrm{crit}} - 1$, depending on whether the chosen score is prediction-based or hyperplane-geometry-based.

9 Concrete new ideas to add to the thesis

9.1 Spectral false nearest neighbors

This is perhaps the most natural bridge to existing methodology. Replace the single thresholded FNN curve by a scale-similarity matrix over dimensions. For example, let

$$G(k) = 1 - F_{\mathrm{FNN}}(k).$$

This is increasing and saturating. Your monotone scale-selection method can detect the dimension where the embedded birth-death stationary distribution is minimized. This gives direct precedence from Kennel–Brown–Abarbanel [4], but your selection rule is new.

9.2 Spectral order selection from residual topology

For each candidate order k, fit an AR(k) model and examine the residual series $\hat{Z}_t^{(k)}$. If $k < p$, residuals retain dependence. If $k \geq p$, residuals should be approximately white. Build delay clouds of residuals and compute topological/geometric dependence scores. The selected order is the smallest k for which residual topology becomes noise-like. This connects the method to classical residual diagnostics, but replaces autocorrelation plots by TDA summaries.

9.3 Bifiltration over dimension and radius

Instead of choosing between k and ε, treat both as parameters:

$$(k, \varepsilon) \mapsto K_{k,\varepsilon}.$$

This is naturally a two-parameter persistence problem. Full multiparameter persistence is mathematically harder, but one can use slices: for each line

$$\varepsilon = ak + b,$$

compute ordinary persistence or your spectral score. This is an ambitious extension but can be presented as future work.

9.4 Zigzag over projected full clouds

Let $P_{k_{\max}}$ be the full delay cloud and let π_k be projection to the first k coordinates. Since projection does not induce inclusion of Vietoris–Rips complexes in a simple way, use the zigzag

$$\mathrm{VR}_\varepsilon(\mathcal{P}_k) o \mathrm{VR}_\varepsilon(\mathcal{P}_k) \cup \mathrm{VR}_\varepsilon(\mathcal{P}_k + 1) \leftarrow \mathrm{VR}_\varepsilon(\mathcal{P}_k + 1).$$

This is a principled topological way to compare changing ambient dimensions, backed by zigzag persistence [7].

9.5 Stability under block bootstrap

Since delay vectors are serially dependent, ordinary bootstrap is inappropriate. Use block bootstrap: resample blocks of the time series, recompute k_{crit}, and report the empirical distribution of $\hat{p}$. A stable method should concentrate around the true order. This gives a practical validation section.

Idea	Promise	Comment
Two-stage ε then k selection	Very high	Easiest to justify and implement; does not force artificial inclusions across dimensions.
Linear-prediction lift	High	Very natural for AR processes; gives interpretable residual and reconstruction scores.
Temporal weighted complex	High	Fixes the issue that the pure consecutive-index complex ignores geometry.
Persistence-diagram kernel over k	Medium-high	Good literature support; less directly connected to your min/meet matrix but elegant.
Zigzag over dimensions	Medium	Mathematically appropriate but may become too much for the thesis if fully developed.
Full two-parameter persistence	Future work	Very interesting but probably too ambitious for the main thesis.
Directed/path homology	Future work	Conceptually attractive for time direction, but adds a large new theory.

Table 1: Assessment of possible extensions.

10 What is most promising?

11 Suggested chapter structure

A clean thesis chapter could be organized as follows.

1. **AR geometry of delay embeddings.** Prove the hyperplane/slab relation for $k = p+1$, the multi-slab relation for $k > p+1$, and the zonoid support statement for bounded innovations.

2. **Why k is a scale.** Explain that k controls the amount of memory visible to the point cloud, analogous to how ε controls geometric resolution.

3. **Candidate constructions.** Present the three routes: separate ε-filtrations for each k, direct kernel over diagrams/signatures, and temporal weighted filtrations.

4. **Recommended pipeline.** Define one concrete algorithm and state what it is expected to recover.

5. **Numerical experiments.** Simulate AR(1), AR(2), AR(3) under different innovations and sample sizes. Plot $G(k)$, $v^*(k)$, $\varepsilon_{\mathrm{crit}}(k)$, and estimated order.

6. **Limitations.** Discuss ARMA contamination, near-unit roots, heavy tails, short samples, and high-dimensional distance concentration.

12 Limitations and cautions

There are several points where the method can fail.

- If the AR coefficients are weak, the geometric transition at p may be subtle.

- If roots are close to the unit circle, the cloud may look almost low-dimensional for reasons unrelated to p.
- If innovations are heavy-tailed, outliers may dominate Vietoris–Rips complexes.
- If T is small, persistent homology and nearest-neighbor statistics may be unstable.
- If $k_{\max}$ is too large, Euclidean distances concentrate. Standardization, PCA preprocessing, or k-nearest-neighbor complexes may be necessary.
- For ARMA(p, q) processes, the finite-order AR geometry is blurred by the MA component. The method may then detect an effective memory scale rather than the exact AR order.

13 Main conclusion

Your proposed application is difficult but feasible if it is framed carefully. The strongest claim to aim for in a master's thesis is not a universal theorem saying that the method always recovers p. A more realistic and valuable contribution is:

> For stationary AR(p) processes, delay-coordinate point clouds undergo a geometric transition around $k = p+1$. This transition can be detected by adapting the spectral scale-selection principle from filtration radii to embedding dimensions, especially when combined with prediction-residual, intrinsic-dimension, or persistence-stability functionals.

This is a coherent and original bridge between classical time-series order selection, delay-coordinate geometry, and your Perron–Frobenius scale-selection framework.

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